

BOBSLEDDING

Tags:

Time Limit: 1 s Memory Limit: 128 MB
 Submission: 67 AC: 32 Score: 99.33

Submit

Codes

AI Code Tips

Description

Dr.Kong has entered a bobsled competition because he hopes his hefty weight will give his an advantage over the L meter course ($2 \leq L \leq 1000$).

Dr.Kong will push off the starting line at 1 meter per second, but his speed can change while he rides along the course. Near the middle of every meter Bessie travels, he can change his speed either by using gravity to accelerate by one meter per second or by braking to stay at the same speed or decrease his speed by one meter per second.

Naturally, Dr.Kong must negotiate N ($1 \leq N \leq 500$) turns on the way down the hill. Turn i is located T_i meters from the course start ($1 \leq T_i \leq L-1$), and he must be enter the corner meter at a peed of at most S_i meters per second ($1 \leq S_i \leq 1000$). Dr.Kong can cross the finish line at any speed he likes.

Help Dr.Kong learn the fastest speed he can attain without exceeding the speed limits on the turns.

Consider this course with the meter markers as integers and the turn speed limits in brackets (e.g., '[3]'):

0 1 2 3 4 5 6 7 [3] 8 9 10 11 [1] 12 13 [8] 14

(Start) |-----?| (Finish)

Below is a chart of Dr.Kong 's speeds at the beginning of each meter length of the course:

Max: [3] [1] [8]

Mtrs: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

Spd: 1 2 3 4 5 5 4 3 4 3 2 1 2 3 4

His maximum speed was 5 near the beginning of meter 4.

Input

Line 1: Two space-separated integers: L and N.

Lines 2..N+1: Line i+1 describes turn i with two space-separated integers: T_i and S_i .

Output

Line 1: A single integer, representing the maximum speed which Dr.Kong can attain between the start and the finish line, inclusive.

Samples

input	Copy
<pre>14 3 7 3 11 1 13 8</pre>	
output	Copy
<pre>5</pre>	

Source

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